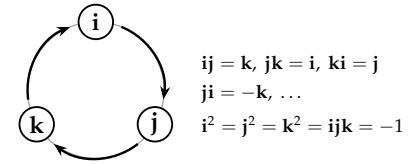


Quaternions

Hamilton's numbers $q = w + xi + yj + zk$: the algebra, rotation as a sandwich $q \mathbf{v} q^*$, conversions, interpolation, kinematics, and the conventions that bite.



1 THE ALGEBRA $\mathbb{H}$

A quaternion is $q = w + xi + yj + zk$ with $w, x, y, z \in \mathbb{R}$. Write it as a scalar and a 3-vector, $q = (s, \mathbf{v})$. Multiplication is associative and distributive but *not* commutative; every $q \neq 0$ has an inverse, so $\mathbb{H}$ is a division algebra.

$$(s_1, \mathbf{v}_1)(s_2, \mathbf{v}_2) = (s_1 s_2 - \mathbf{v}_1 \cdot \mathbf{v}_2, s_1 \mathbf{v}_2 + s_2 \mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2)$$

$$q^* = (s, -\mathbf{v}), \quad |q|^2 = qq^* = s^2 + |\mathbf{v}|^2, \quad q^{-1} = q^* / |q|^2$$

$\times$	1	i	j	k
1	1	i	j	k
i	i	-1	k	-j
j	j	-k	-1	i
k	k	j	-i	-1

row $\times$ column shaded entries are negative
 $ji = -ij$

- Pure quaternions ($s = 0$) are 3-vectors: $\mathbf{u}\mathbf{v} = -\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \times \mathbf{v}$, so $\mathbf{u}^2 = -|\mathbf{u}|^2$.
- Commutator: $q_1 q_2 - q_2 q_1 = 2(0, \mathbf{v}_1 \times \mathbf{v}_2)$. Two quaternions commute iff their vector parts are parallel.
- $(pq)^* = q^* p^*$, $(pq)^{-1} = q^{-1} p^{-1}$, $|pq| = |p| |q|$.
- Norm is multiplicative, which is Euler's four-square identity: a product of two sums of four squares is a sum of four squares.
- Real part: $\text{Re } q = \frac{1}{2}(q + q^*)$; $\text{Re}(pq) = \text{Re}(qp)$; $\text{Re}(pq^*) = p \cdot q$ (the 4D dot product).

Polar form. Any $q \neq 0$ is $q = |q|(\cos \alpha + \mathbf{u} \sin \alpha)$ with $|\mathbf{u}| = 1$, $\mathbf{u}^2 = -1$, $\alpha \in [0, \pi]$. Since $\mathbf{u}$ behaves like i , De Moivre holds: $q^t = |q|^t (\cos t\alpha + \mathbf{u} \sin t\alpha)$.

$$e^{\alpha \mathbf{u}} = \cos \alpha + \mathbf{u} \sin \alpha, \quad \log q = \ln |q| + \mathbf{u} \operatorname{atan2}(|\mathbf{v}|, s)$$

Caution: $e^p e^q \neq e^{p+q}$ unless p, q commute.

Matrix pictures. Left and right multiplication are linear maps of $\mathbb{R}^4$ (vectors ordered w, x, y, z):

$$L(q) = \begin{pmatrix} w & -x & -y & -z \\ x & w & -z & y \\ y & z & w & -x \\ z & -y & x & w \end{pmatrix}, \quad R(q) = \begin{pmatrix} w & -x & -y & -z \\ x & w & z & -y \\ y & -z & w & x \\ z & y & -x & w \end{pmatrix}$$

with $pq = L(p)q = R(q)p$; for unit q both are orthogonal, and L and R always commute. As complex 2×2 matrices,

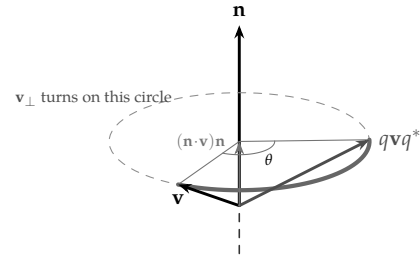
$$w + xi + yj + zk \mapsto \begin{pmatrix} w + xi & y + zi \\ -y + zi & w - xi \end{pmatrix},$$

a ring homomorphism with $\det = |q|^2$. Unit quaternions become exactly $SU(2)$, and $i, j, k \mapsto i\sigma_z, i\sigma_y, i\sigma_x$ (Pauli matrices).

2 ROTATION IS A SANDWICH

$$q = \cos \frac{\theta}{2} + \mathbf{n} \sin \frac{\theta}{2} \quad (|\mathbf{n}| = 1) \implies \mathbf{v}' = q \mathbf{v} q^*$$

rotates $\mathbf{v}$ by θ about $\mathbf{n}$, anticlockwise when $\mathbf{n}$ points at you (right-hand rule).

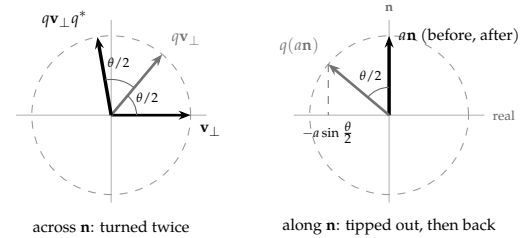


The component along $\mathbf{n}$ is fixed; the component $\mathbf{v}_\perp$ across $\mathbf{n}$ turns through θ on a circle of radius $|\mathbf{v}_\perp|$.

Expanding the sandwich gives Rodrigues' formula:

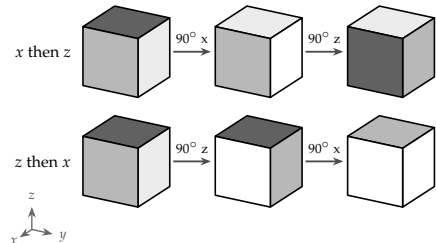
$$q \mathbf{v} q^* = \mathbf{v} \cos \theta + (\mathbf{n} \times \mathbf{v}) \sin \theta + \mathbf{n}(\mathbf{n} \cdot \mathbf{v})(1 - \cos \theta).$$

Why half the angle? Split $\mathbf{v} = a\mathbf{n} + \mathbf{v}_\perp$. Left multiplication alone does two things:



$q\mathbf{v}$ turns $\mathbf{v}_\perp$ by $\theta/2$ but gives $a\mathbf{n}$ a real part $-a \sin \frac{\theta}{2}$, so $q\mathbf{v}$ is not a vector. Multiplying by q^* on the right turns $\mathbf{v}_\perp$ by the other $\theta/2$ and cancels the real part.

Composition. Apply q_1 then q_2 : $q = q_2 q_1$, since $q_2(q_1 \mathbf{v} q_1^*) q_2^* = (q_2 q_1) \mathbf{v} (q_2 q_1)^*$. Same order as matrices, $R(q_2 q_1) = R(q_2) R(q_1)$.



The same two quarter turns in the two orders leave the block in different orientations: $q_z q_x = \frac{1}{2}(1 + i + j + k)$, but $q_x q_z = \frac{1}{2}(1 + i - j + k)$. Their difference is the commutator $q_z q_x - q_x q_z = 2(0, \mathbf{v}_z \times \mathbf{v}_x) = j$.

Example 1 90° about x , then 90° about z . $q_1 = \frac{1}{\sqrt{2}}(1 + i)$, $q_2 = \frac{1}{\sqrt{2}}(1 + k)$,

$$q_2 q_1 = \frac{1}{2}(1 + i + k + ki) = \frac{1}{2}(1 + i + j + k).$$

So $\cos \frac{\theta}{2} = \frac{1}{2}$: a 120° turn about $(1, 1, 1)/\sqrt{3}$, which cycles the axes $x \rightarrow y \rightarrow z \rightarrow x$.

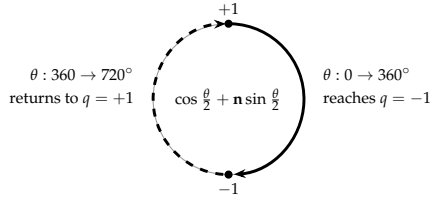
Rotating many vectors fast. With $q = (w, \mathbf{u})$ unit:

$$\mathbf{t} = 2\mathbf{u} \times \mathbf{v}, \quad \mathbf{v}' = \mathbf{v} + w\mathbf{t} + \mathbf{u} \times \mathbf{t}$$

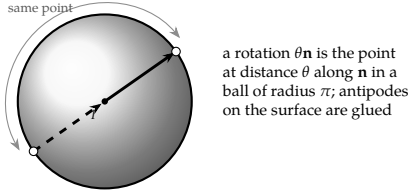
(18 multiplications, 12 additions). For thousands of vectors, convert to a matrix once (9 and 6 per vector).

3 THE DOUBLE COVER

Unit quaternions form the 3-sphere $S^3 \subset \mathbb{R}^4$, a group isomorphic to $SU(2)$. The sandwich map $S^3 \rightarrow SO(3)$ is onto and exactly two-to-one: q and $-q$ give the same rotation, since the signs cancel.



Turning steadily about a fixed axis: the rotation is back after 360° , the quaternion only after 720° . The loop of rotations through 360° cannot be shrunk to a point; the 720° loop can (the plate or belt trick), so $\pi_1(SO(3)) = \mathbb{Z}/2$.



$SO(3)$ as a ball with opposite surface points glued ($\mathbb{RP}^3$). The 360° turn is a loop that leaves through $\pi\mathbf{n}$ and returns through $-\pi\mathbf{n}$; it cannot be pulled inside the ball. Going round twice, the two crossings can be slid together and cancelled. S^3 is the double cover that unglues the surface.

- Angle between orientations p, q : $\theta = 2 \arccos |p \cdot q|$ (the absolute value handles the sign). Near 0 use $2 \operatorname{atan2}(|\operatorname{vec}(p^*q)|, |\operatorname{Re}(p^*q)|)$ for accuracy.
- Uniform random rotation (Shoemake): for $u_1, u_2, u_3 \sim U[0, 1]$,

$$q = (\sqrt{1-u_1} \sin 2\pi u_2, \sqrt{1-u_1} \cos 2\pi u_2, \sqrt{u_1} \sin 2\pi u_3, \sqrt{u_1} \cos 2\pi u_3).$$

Normalising a 4D Gaussian vector also works.

- Haar measure on $SO(3)$ is the uniform measure on S^3 ; the rotation angle then has density $\frac{1}{\pi}(1 - \cos \theta)$ on $[0, \pi]$.

4 CONVERSIONS

Axis-angle. $q = (\cos \frac{\theta}{2}, \mathbf{n} \sin \frac{\theta}{2})$; back: $\theta = 2 \operatorname{atan2}(|\mathbf{v}|, s)$, $\mathbf{n} = \mathbf{v}/|\mathbf{v}|$ (any $\mathbf{n}$ if $\mathbf{v} = 0$).

To a matrix (unit $q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$):

$$R = \begin{pmatrix} 1-2(y^2+z^2) & 2(xy-wz) & 2(xz+wy) \\ 2(xy+wz) & 1-2(x^2+z^2) & 2(yz-wx) \\ 2(xz-wy) & 2(yz+wx) & 1-2(x^2+y^2) \end{pmatrix}$$

Trace: $\operatorname{tr} R = 4w^2 - 1 = 1 + 2 \cos \theta$.

From a matrix (Shepperd). Compute the four candidates

$$4w^2 = 1 + \operatorname{tr} R, \quad 4x^2 = 1 + R_{11} - R_{22} - R_{33},$$

and likewise $4y^2, 4z^2$. Take the *largest*, then read the others from the off-diagonal sums and differences, e.g. if x is largest:

$$w = \frac{R_{32}-R_{23}}{4x}, \quad y = \frac{R_{12}+R_{21}}{4x}, \quad z = \frac{R_{13}+R_{31}}{4x}.$$

Trap. The textbook route $w = \frac{1}{2}\sqrt{1 + \operatorname{tr} R}$ then $x = (R_{32} - R_{23})/4w$ divides by $w \approx 0$ near a half turn. For 180° about $(1, 1, 0)/\sqrt{2}$, $\operatorname{tr} R = -1$ and $w = 0$ exactly; Shepperd picks $4x^2 = 2$ and returns $q = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$.

Euler angles (aerospace z - y - x : $R = R_z(\psi)R_y(\theta)R_x(\varphi)$, yaw, pitch, roll). With $c_\bullet = \cos \frac{\bullet}{2}$, $s_\bullet = \sin \frac{\bullet}{2}$:

$$\begin{aligned} w &= c_\varphi c_\theta c_\psi + s_\varphi s_\theta s_\psi, & x &= s_\varphi c_\theta c_\psi - c_\varphi s_\theta s_\psi, \\ y &= c_\varphi s_\theta c_\psi + s_\varphi c_\theta s_\psi, & z &= c_\varphi c_\theta s_\psi - s_\varphi s_\theta s_\psi. \end{aligned}$$

Back:

$$\begin{aligned} \varphi &= \operatorname{atan2}(2(wx + yz), 1 - 2(x^2 + y^2)), \\ \theta &= \arcsin(2(wy - zx)), \\ \psi &= \operatorname{atan2}(2(wz + xy), 1 - 2(y^2 + z^2)). \end{aligned}$$

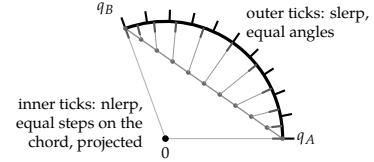
Clamp the arcsine argument to $[-1, 1]$. At $\theta = \pm 90^\circ$ only $\psi \mp \varphi$ is determined (gimbal lock, §6).

Example 2 $q = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$ from Example 1: $\varphi = \operatorname{atan2}(1, 0) = 90^\circ$, $\theta = \arcsin 0 = 0$, $\psi = 90^\circ$. Check: $R_z(90^\circ)R_x(90^\circ)$ sends $x \mapsto y$ and $y \mapsto z$.

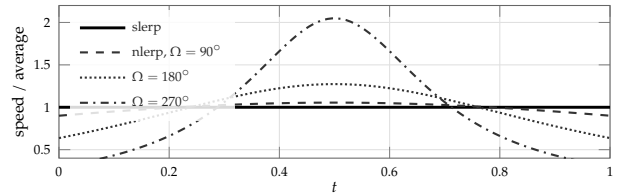
5 INTERPOLATION

Orientations live on S^3 with antipodes identified. The geodesic from q_A to q_B is a great-circle arc; with $\cos \varphi = q_A \cdot q_B$ (after flipping $q_B \rightarrow -q_B$ if negative), the rotation turned through is $\Omega = 2\varphi$.

$$\operatorname{slerp}(t) = \frac{\sin(1-t)\varphi}{\sin \varphi} q_A + \frac{\sin t\varphi}{\sin \varphi} q_B = q_A (q_A^* q_B)^t$$



The plane through $0, q_A, q_B$ in $\mathbb{R}^4$. nlerp , $\operatorname{normalise}((1-t)q_A + tq_B)$, follows the same arc but its ticks bunch at the ends.



nlerp 's peak speed, at $t = \frac{1}{2}$, is $2 \tan(\varphi/2)/\varphi$ times the average: 1.05, 1.27, 2.05 for the three curves. Fine for small steps, poor for large ones.

- Always flip** $q_B \rightarrow -q_B$ when $q_A \cdot q_B < 0$, or the path goes the long way, $360^\circ - \Omega$.
- When φ is tiny, $\sin \varphi \rightarrow 0$: switch to nlerp below $\varphi \approx 10^{-3}$.
- Interpolating Euler angles is not a geodesic, and an angle crossing $\pm 180^\circ$ unwinds the long way round.
- Smooth splines through many keys: *squad*, or De Casteljau with slerp in place of lerp ; both are C^1 on S^3 .

Example 3 Halfway from 1 to $q = \frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$: $\cos \varphi = \frac{1}{2}$, $\varphi = 60^\circ$, and slerp at $t = \frac{1}{2}$ is $(1+q)/[1+|q|] = (\frac{\sqrt{3}}{2}, \frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}})$: a 60° turn about $(1, 1, 1)/\sqrt{3}$, half of the 120° .

6 KINEMATICS AND GIMBAL LOCK

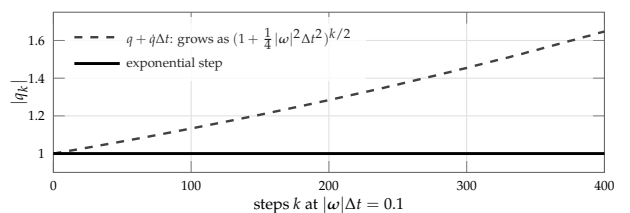
$$\dot{q} = \frac{1}{2} \omega q \quad (\omega \text{ in the world frame})$$

$$\dot{q} = \frac{1}{2} q \omega_b \quad (\omega_b \text{ in the body frame})$$

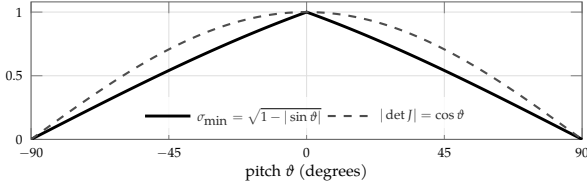
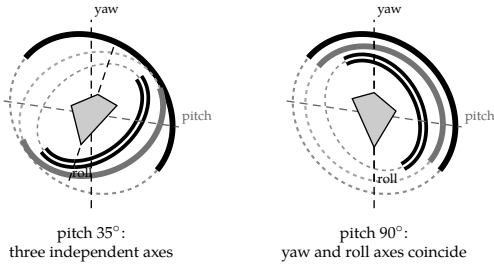
Conversely $\omega = 2 \dot{q} q^*$ and $\omega_b = 2 q^* \dot{q}$. The exact step for constant body rate over Δt is

$$q_{k+1} = q_k \exp(\frac{1}{2} \omega_b \Delta t) = q_k \left(\cos \frac{|\omega_b| \Delta t}{2} + \hat{\omega}_b \sin \frac{|\omega_b| \Delta t}{2} \right),$$

which stays on S^3 . A plain Euler step $q + \dot{q} \Delta t$ drifts off it: renormalise every step.



Gimbal lock. Euler rates map to angular velocity by $\omega = \dot{\psi} \hat{z} + \dot{\theta} R_z \hat{y} + \dot{\varphi} R_z R_y \hat{x} = J(\psi, \theta, \varphi)^T$, with $|\det J| = \cos \theta$ and singular values $1, \sqrt{1 \pm \sin \theta}$.



At $\theta = \pm 90^\circ$ the roll axis lies on the yaw axis: one direction of turning is unreachable, and near it Euler rates of order $1/\sigma_{\min}$ are needed. Every three-parameter chart of $SO(3)$ has such a point; the quaternion's four numbers with one constraint have none.

7 REPRESENTATIONS COMPARED

	numbers / constraint	singular?	compose / interpolate
Euler angles	3 / none	yes, $\theta = \pm 90^\circ$	via matrix / poor
axis-angle	3 or 4 / $ \mathbf{n} = 1$	axis undefined at $\theta = 0$	awkward / along one axis only
matrix	9 / $R^T R = I, \det = 1$	no	27 mult / no
quaternion	4 / $ q = 1$	no; $\pm q$ double	16 mult / slerp

8 CONVENTIONS THAT BITE

- **Storage order.** (w, x, y, z) in most mathematics and MATLAB; Eigen's constructor takes (w, x, y, z) but stores (x, y, z, w) ; ROS and SciPy use (x, y, z, w) . Read the documentation, never the variable names.
- **Hamilton vs JPL.** JPL/Shuster quaternions use $\mathbf{ij} = -\mathbf{k}$, which reverses the order of composition. Aerospace and robotics papers mix both.
- **Active vs passive.** $q\mathbf{v}q^*$ rotates the vector; $q^*\mathbf{v}q$ gives the same vector's coordinates in the rotated frame. A sign error here looks like a transposed matrix.
- **Local vs global increments.** $q\delta q$ applies δq about body axes; $\delta q q$ about world axes.
- **Sign continuity.** A stream of orientations (sensor output, keyframes) can flip sign between samples; before filtering or

differencing, flip each q_k so that $q_{k-1} \cdot q_k \geq 0$.

- **Drift.** Products of unit quaternions lose unit length slowly in floating point; renormalise, which is far cheaper than re-orthogonalising a matrix.

9 RECIPES

Shortest rotation taking $\mathbf{u}$ to $\mathbf{v}$ (both unit):

$$q = \text{normalise}(1 + \mathbf{u} \cdot \mathbf{v}, \mathbf{u} \times \mathbf{v}).$$

This is the half-angle trick in reverse: $(1 + \cos \theta, \mathbf{n} \sin \theta)$ has argument $\theta/2$. If $\mathbf{u} \cdot \mathbf{v} \approx -1$, use $q = (0, \mathbf{n})$ for any unit $\mathbf{n} \perp \mathbf{u}$.

Swing-twist about a unit axis $\mathbf{d}$. For $q = (w, \mathbf{v})$:

$$q_{\text{tw}} = \text{normalise}(w, (\mathbf{v} \cdot \mathbf{d}) \mathbf{d}), \quad q_{\text{sw}} = q q_{\text{tw}}^*,$$

so $q = q_{\text{sw}} q_{\text{tw}}$: a twist about $\mathbf{d}$ followed by a swing about an axis perpendicular to $\mathbf{d}$. Used for joint limits and camera roll.

Small rotations. $q(\delta \theta) \approx (1, \frac{1}{2} \delta \theta)$, $R \approx I + [\delta \theta]_{\times}$, and small rotations commute to first order. Error-state filters store the error as $\delta \theta \approx 2 \text{vec}(\hat{q}^* q)$.

Averaging orientations. Averaging components fails because of the sign ambiguity. Take the unit eigenvector of the largest eigenvalue of $M = \sum_i w_i q_i q_i^T$ (4×4), which is indifferent to the sign of each q_i (Markley et al. 2007).

Quaternions worth knowing by sight.

rotation	q
identity	1 (or -1)
90° about x (so for y, z)	$\frac{1}{\sqrt{2}}(1 + \mathbf{i})$
180° about x, y, z	$\mathbf{i}, \mathbf{j}, \mathbf{k}$
180° about unit $\mathbf{n}$	$\mathbf{n}$ itself
120° about $(1, 1, 1)/\sqrt{3}$	$\frac{1}{2}(1 + \mathbf{i} + \mathbf{j} + \mathbf{k})$
θ about z	$\cos \frac{\theta}{2} + \mathbf{k} \sin \frac{\theta}{2}$

10 CONTEXT

Hamilton found the rules on 16 October 1843, walking along the Royal Canal in Dublin, and cut $i^2 = j^2 = k^2 = ijk = -1$ into Broom Bridge. Frobenius (1878): $\mathbb{R}, \mathbb{C}$ and $\mathbb{H}$ are the only finite-dimensional associative real division algebras. Doubling once more (Cayley–Dickson) gives the octonions, which lose associativity; Hurwitz: $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ are the only normed ones. Pairs of unit quaternions act on $\mathbb{R}^4$ by $x \mapsto pxq$, giving $S^3 \times S^3 \rightarrow SO(4)$, again two-to-one.